

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-LV-6701-01 Sheet 1 of 1
OPL Title	Control Valve Stroke Calibration (DVC6200)	Discipline	Instrument	
Equipment	LV-6701 - SEPARATOR LEVEL CONTROL VALVE	Area / Unit	6700 - LP SEPARATOR SYSTEM	
Related Interlock	SEQ-6701 (SIL 1)	P&ID Ref	TJC-LLD-PID-6701	
Classification:	☒ Basic Knowledge	☐ Improvement	☐ Trouble Case	Aspect: P-Q-C-D- S -M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **control valve stroke calibration (dvc6200)** for LV-6701 (SEPARATOR LEVEL CONTROL VALVE) in the LP SEPARATOR SYSTEM. Correct mapping of the 4-20 mA signal to 0-100 % valve stroke is essential for stable separator level control. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **LOW CRITICAL** handling **Hexane + polymer bottoms** at up to ASME 300# / 120 degC. Observe the following before and during the task:

- De-energise and isolate the loop; place the function to bypass/inhibit under permit.
- Inform the panel operator before working on any trip, valve or transmitter.
- Take care with instrument air / hydraulic pressure when disconnecting tubing.
- Verify intrinsic-safety / Ex integrity is restored before returning to service.
- Wear correct PPE and follow ESD/override control procedures.

3. TOOLS & MATERIALS REQUIRED

- HART communicator / positioner software (e.g. ValveLink)
- Loop calibrator & decade box / pressure source
- Digital multimeter and clamp meter
- Span/zero test gas or reference standards
- Torque screwdriver and small hand tools
- Certified test/inhibit permit

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Connect ValveLink to the DVC6200 positioner	Loop isolated / under permit
2	Run the auto-calibration over the full stroke	Reference standard traceable
3	Verify 4 mA = closed and 20 mA = 38 mm fully open	As-found & as-left recorded
4	Check hysteresis and dead-band are within 1 %	Trip / control action verified
5	Save the calibration and record the as-left data	Return to service confirmed

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
LV-6701 not responding to LIC-6701, level control lost	Stem seized in PTFE packing (over-tightened + fines)	Freed stem, renewed packing, lubricated, re-stroked and calibrated
Positioner giving position error after drift	Positioner feedback out of calibration	Recalibrated DVC6200, updated firmware, verified travel
Hexane weeping at LV-6701 stem packing	PTFE V-ring packing worn	Renewed PTFE V-ring packing set, re-torqued gland

6. KEY LEARNING POINTS

- Correct mapping of the 4-20 mA signal to 0-100 % valve stroke is essential for stable separator level control.
- Respect the interlock SEQ-6701: e.g. LSHH-6710 trips at > 85 % - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-6701 before working on LV-6701.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

Prepared by	Reviewed by (Supervisor)	Approved by (Manager)	Date of Sharing
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